

MODULE 2**Cybercrime and Cyber law****SYLLABUS:**

Classification of cyber crimes, Common cyber crimes- cyber crime targeting computers and mobiles, cyber crime against women and children, financial frauds, social engineering attacks, malware and ransomware attacks, zero day and zero click attacks, Cybercriminals modus-operandi, Reporting of cyber crimes, Remedial and mitigation measures, Legal perspective of cyber crime, IT Act 2000 and its amendments, Cyber crime and offences

2.0 Cyber crimes

Cyber crimes refer to illegal activities that are carried out using computers, digital devices, or networks, primarily targeting information systems, digital data or online users. As the world becomes increasingly interconnected through the internet and digital technologies, cyber crimes have emerged as one of the most serious threats to individuals, businesses, and governments alike. These crimes range from relatively simple offenses like email fraud and identity theft to more complex attacks such as hacking, ransomware attacks, financial frauds, cyber terrorism, and data breaches. One of the most troubling aspects of cyber crime is that it can be committed remotely, anonymously, and on a global scale, making investigation and prosecution more challenging.

Cyber criminals use a variety of tools and techniques including malware, phishing, social engineering, and denial-of-service (DoS) attacks to exploit system vulnerabilities and human error. For example, phishing emails may trick users into revealing sensitive information such as passwords or credit card numbers, while advanced persistent threats (APTs) may stealthily infiltrate networks to exfiltrate confidential corporate or government data. Some

cyber crimes are financially motivated, while others may aim to disrupt infrastructure, spread misinformation, or conduct surveillance.

Governments and international agencies have recognized the growing threat posed by cyber crimes and are establishing cyber laws, digital forensics units, and cybercrime cells to combat these activities. Public awareness, cyber hygiene, strong authentication methods, and timely software updates are essential to minimizing the risk of becoming a victim. As digital dependency grows, understanding, detecting, and preventing cyber crimes has become an urgent priority for maintaining trust, safety, and resilience in the digital age.

2.0.1 What are cyber crimes

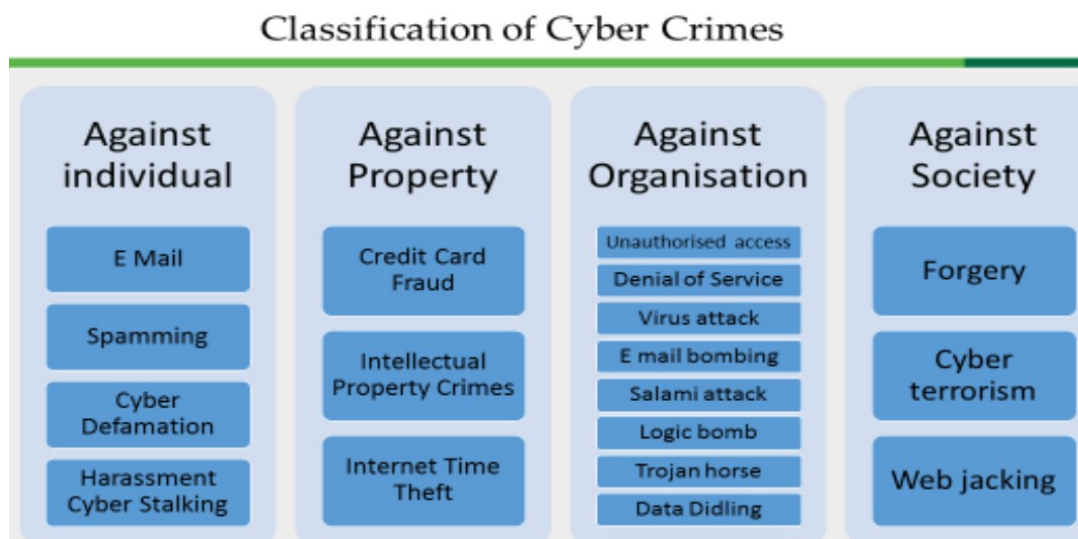
Cyber crime refers to any unlawful activity carried out using computers, computer networks, or the Internet, where digital systems, electronic data, or online services are misused, attacked or exploited. In such crimes, the computer may act as the **target of the crime**, the **tool used to commit the crime** or **both**. Cyber crimes include unauthorized access to computer systems, data theft, online fraud, identity theft, spreading malware and disruption of electronic services. These crimes threaten the confidentiality, integrity and availability of information and have a serious impact on individuals, organizations, and electronic commerce systems.

- These crimes are also commonly referred to as computer crimes or internet crimes.
- In cyber crimes, the computer may act as a tool, a target or both tool and target of the criminal activity.
- The primary objective of committing cyber crimes is to access, steal, manipulate or destroy confidential information.
- Cyber criminals commit such crimes for monetary gain, political advantage, personal revenge or malicious intent.
- Cyber crimes pose serious threats to data security, privacy, electronic commerce and online trust.

2.1 Classification of cyber crimes

Cyber crimes can be broadly classified into four major categories based on the nature of the victim affected. These include cyber crimes against individuals, such as email abuse,

spamming, cyber defamation, and cyber stalking; cyber crimes against property, which involve credit card fraud, intellectual property crimes, and internet time theft; cyber crimes against organizations, including unauthorized access, denial-of-service attacks, virus attacks, email bombing, logic bombs, Trojan horses, and data diddling; and cyber crimes against society, which include forgery, cyber terrorism, and web jacking. This classification helps in understanding the scope and impact of cyber crimes from legal and security perspectives. Fig. 2.1 illustrates the classification of cyber crimes based on the nature of the target, including individuals, property, organizations, society and government.



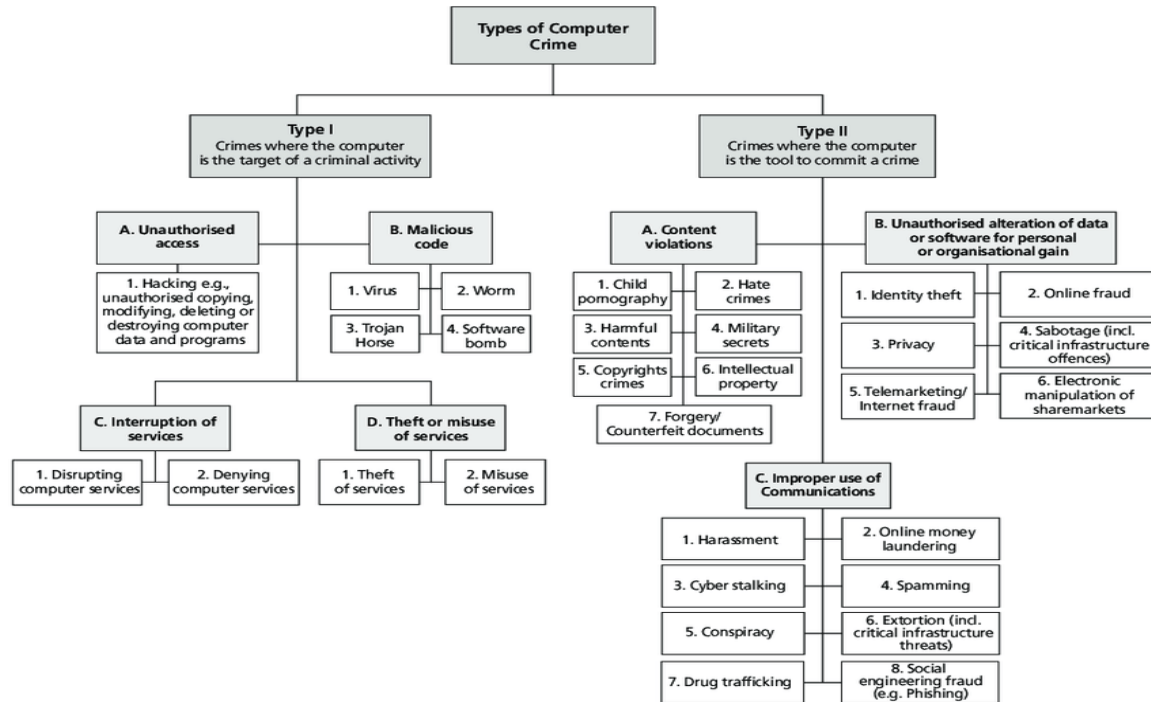


Fig. 2.1 Classification of Cyber Crimes

2.1 Classifying cybercrimes-broad and narrow

	Cybercrime in Narrow Sense	Cybercrime in Broad Sense	
Role of computer	Computer as an object The computer / information stored on the computer is the subject/target of the crime	Computer as a tool The computer/or information stored on the computer constitutes an important tool for committing the crime	Computer as the environment or context The computer /information stored on the computer play a non-substantial role in the act of crime, but does contain evidence of the crime
Examples	Hacking, sabotage, virtual child pornography	Computer fraud, forgery distribution of child pornography	Murder using computer techniques, bank robbery and drugs trade

cyber crimes can be classified under three heads, depending on the groups they are targeted at. Table 2.1 illustrates the broad and narrow classification of cyber crimes.

2.1.1. Cyber crime against Individual

These crimes directly affect an individual's **privacy, reputation or personal safety**. They include crimes such as email abuse, spamming, cyber defamation, cyber stalking and online harassment. Cyber crimes against individuals are offences that violate a person's **privacy, dignity, mental peace or financial security**. These crimes are committed by misusing electronic communication and internet services. The main objective of such crimes is to threaten, defame, mentally harass or illegally exploit individuals through electronic means.

- **Email spoofing:** A spoofed email is one in which the e-mail header is forged so that the mail appears to originate from one source but actually has been sent from another source.
- **Spamming:** Spamming means sending multiple copies of unsolicited mails or mass e-mails such as chain letters.
- **Cyber Defamation:** This occurs when defamation takes place with the help of computers and/or the Internet. E.g. someone publishes defamatory matter about someone on a website or sends e-mails containing defamatory information.
- **Harassment & Cyber stalking:** Cyber Stalking Means following an individual's activity over internet. It can be done with the help of many protocols available such as e- mail, chat rooms, user net groups.

Fig. 2.2 Cyber Crimes Against Persons depicts offenses such as cyber stalking, online harassment, identity misuse and defamation that harm an individual's privacy, dignity and personal security.



Fig. 2.2 Cyber Crimes Against Persons

2.1.2. Cyber crime Against Property

These crimes involve the **theft, damage or misuse of digital or intellectual property**. Cyber criminals target valuable information and electronic assets for financial gain. Examples include credit card fraud, intellectual property crimes such as software piracy and copyright infringement, and internet time theft. These crimes result in economic loss and violate ownership rights over digital resources.

- **Credit Card Fraud:** As the name suggests, this is a fraud that happens by the use of a credit card. This generally happens if someone gets to know the card number or the card gets stolen.
- **Intellectual Property crimes:** These include Software piracy: Illegal copying of programs, distribution of copies of software. Copyright infringement: Using copyrighted material without proper permission. Trademarks violations: Using trademarks and associated rights without permission of the actual holder. Theft of computer source code: Stealing, destroying or misusing the source code of a computer.
- **Internet time theft:** This happens by the usage of the Internet hours by an unauthorized person which is actually paid by another person.

Fig. 2.3 Cyber Crimes Against Property illustrates illegal activities such as hacking, data theft, identity theft and online fraud that target digital assets, confidential information, and financial resources of individuals and organizations.

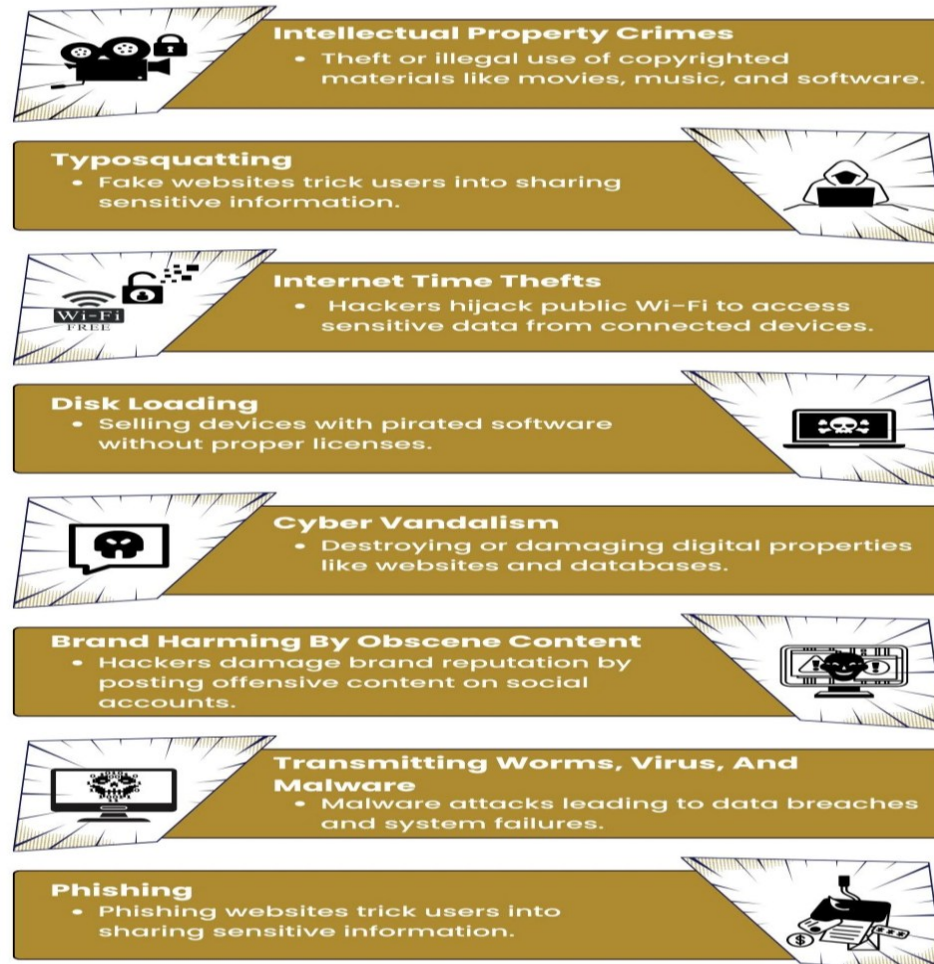


Fig. 2.3 Cyber Crimes Against Property

2.1.3 Cyber crime Against Organization

Cyber crimes against organizations are unlawful activities in which **business firms, government departments, educational institutions or corporate bodies** are targeted using computers and network technologies. These crimes are committed with the intention of **stealing sensitive information, disrupting operations, causing financial loss or damaging the reputation of the organization**. As explained in standard cyber security

textbooks, organizations are prime targets because they store large volumes of valuable data and depend heavily on information systems.

- **Unauthorized Accessing of Computer:** Accessing the computer/network without permission from the owner. It can be of 2 forms: a) Changing/deleting data: Unauthorized changing of data. b) Computer **voyeur:** The criminal reads or copies confidential or proprietary information, but the data is neither deleted nor changed.
- **Denial Of Service:** When Internet server is flooded with continuous bogus requests so as to denying legitimate users to use the server or to crash the server.
- **Computer contamination / Virus attack:** A computer virus is a computer program that can infect other computer programs by modifying them in such a way as to include a (possibly evolved) copy of it. Viruses can be file infecting or affecting boot sector of the computer. Worms, unlike viruses do not need the host to attach themselves to.
- **Email Bombing:** Sending large numbers of mails to the individual or company or mail servers thereby ultimately resulting into crashing.
- **Salami Attack:** When negligible amounts are removed & accumulated in to something larger. These attacks are used for the commission of financial crimes.
- **Logic Bomb:** It is an event dependent program. As soon as the designated event occurs, it crashes the computer, release a virus or any other harmful possibilities.
- **Trojan Horse:** This is an unauthorized program which functions from inside what seems to be an authorized program, thereby concealing what it is actually doing.
- **Data diddling:** This kind of an attack involves altering raw data just before it is processed by a computer and then changing it back after the processing is completed.

2.1.4 Cyber crime Against Society

Cyber crimes against society are offences that affect the **public at large, social order, and national security**. Unlike crimes against individuals or property, these crimes have a **widespread impact** and can disturb social harmony, public confidence and the functioning

of government or public systems. As described in cyber security textbooks, such crimes are considered highly serious due to their large-scale consequences.

One major cyber crime against society is **cyber forgery**, where electronic records, documents, certificates, or digital signatures are illegally created or altered. This can lead to serious misuse of government records, legal documents, and financial data. Another severe form is **cyber terrorism**, which involves the use of computers and networks to threaten national security, spread fear, disrupt critical infrastructure, or influence government policies. These attacks may target defense systems, power grids, banking networks or communication systems.

- **Forgery** : Currency notes, revenue stamps, mark sheets etc. can be forged using computers and high quality scanners and printers.
- **Cyber Terrorism** : Use of computer resources to intimidate or coerce people and carry out the activities of terrorism.
- **Web Jacking** : Hackers gain access and control over the website of another, even they change the content of website for fulfilling political objective or for money.

1. Key Cyber Threats for Organizations in 2026

- **AI-Powered & Agentic Attacks:** Attackers use "agentic AI" to perform autonomous reconnaissance, exploit vulnerabilities in minutes and generate malware variants that can evolve in real-time to evade detection.
- **Deepfake Fraud & Digital Impersonation:** Sophisticated deepfake audio and video are used to impersonate CEOs or high-level executives to authorize fraudulent financial transfers, costing businesses billions globally.
- **Ransomware 5.0 (Multi-Extortion):** Ransomware has moved beyond mere encryption to include data exfiltration, public extortion, and harassment of an organization's clients or suppliers to maximize leverage.
- **Supply Chain & Third-Party Vulnerabilities:** Instead of attacking a well-defended enterprise directly, criminals target weaker third-party vendors, managed service providers (MSPs) or shared cloud code libraries to gain "backdoor" access.

- **Quantum "Harvest Now, Decrypt Later":** Adversaries are actively harvesting encrypted sensitive data today with the intention of decrypting it once quantum computing matures enough to break current encryption standards.
- **Cloud Misconfigurations & Shadow IT:** With widespread cloud adoption, unmanaged "shadow IT" and misconfigured cloud storage remain leading causes of massive data exposure.

2.2 Common cyber crimes

Common cyber crimes include offences directed at computing devices and smartphones through unauthorized access, data manipulation or service disruption, along with digitally enabled abuse affecting women and children such as online harassment, exploitation, and circulation of harmful content. Illicit monetary activities arise through deceptive transactions, payment manipulation, and misuse of electronic banking platforms, while social engineering relies on psychological tactics to mislead users into revealing confidential details. Threats involving malicious software encrypt, damage or covertly control systems, and advanced attack techniques exploit previously unknown weaknesses or operate without user interaction, making detection and prevention increasingly challenging. Common cyber crimes represent frequently occurring digital offences that exploit networked systems, online platforms and human vulnerabilities for illegal gain or harm, and include the following:

2.2.1 Cyber crime targeting computers and mobiles

Cyber crimes targeting computers and mobile devices refer to unlawful activities in which **computers, smartphones, tablets and other digital devices themselves become the direct targets of attack**. As explained in cyber security textbooks like **Belapure and Godbole**, these crimes aim to **damage devices, steal stored data, gain unauthorized control or disrupt normal functioning** of the system.

In the case of **computers**, cyber criminals use techniques such as **virus attacks, worms, Trojan horses, logic bombs, spyware and ransomware** to corrupt files, destroy data or take control of the system. **Hacking and unauthorized access** are also common, where attackers break into computer systems to steal confidential information or misuse system resources.

With the widespread use of smartphones, **mobile devices have become major targets of cyber crimes**. Mobile cyber crimes include **mobile malware, malicious apps, SMS phishing (smishing), Bluetooth attacks and spyware**, which can steal personal data, banking details, contacts and messages. Attackers may also remotely control mobile devices or misuse them for illegal activities.

Overall, cyber crimes targeting computers and mobiles pose serious threats to **data security, privacy and system integrity**. These crimes highlight the need for strong security measures such as antivirus software, secure applications, regular updates, and user awareness to protect digital devices from cyber attacks.

- Cybercrime targeting computers and mobile devices is a growing concern in today's digital world.
- These crimes encompass a wide range of illegal activities conducted using technology, often with the goal of financial gain, data theft, or causing harm to individuals, organizations, or governments.
- Here are some common types of cybercrimes that target computers and mobiles:
 1. **Malware Attacks:** Malicious software (malware) is designed to infect computers and mobile devices. This includes viruses, worms, Trojans, ransomware, spyware and adware. Malware can steal data, damage systems, or hold data hostage for a ransom.
 2. **Phishing:** Phishing attacks involve tricking individuals into revealing sensitive information like passwords, credit card numbers, or personal details by posing as a legitimate entity through email, text messages, or fake websites.
 3. **Identity Theft:** Cybercriminals can steal personal information, such as Social Security numbers and financial data, to commit fraud, open accounts in victims' names, or access their financial resources.
 4. **Online Scams:** Various online scams target individuals, such as advance-fee fraud, lottery scams, and romance scams. These scams deceive people into sending money or personal information to fraudsters.

5. **DDoS Attacks:** Distributed Denial of Service (DDoS) attacks overwhelm a target's computer or network with traffic, making it unavailable to users. These attacks are often used to disrupt services or extort money.
 6. **Data Breaches:** Cybercriminals infiltrate organizations to steal sensitive data like customer information, trade secrets or financial records. These breaches can result in significant financial losses and reputational damage.
 7. **Cyberbullying:** Cyberbullying involves the use of technology to harass, threaten or intimidate individuals. It can take place through social media, messaging apps or email.
 8. **Mobile Device Theft and Hacking:** Criminals can steal mobile devices for resale or hack into them to access personal data, financial information or install malware.
 9. **Cyber Extortion:** Criminals may threaten to release sensitive or embarrassing information unless a victim pays a ransom. This can involve sextortion (threatening to expose explicit content) or other forms of extortion.
 10. **Insider Threats:** Employees or individuals with insider access to computer systems and data may misuse their privileges to steal or manipulate information.
 11. **Cryptojacking:** Cybercriminals use a victim's computer or mobile device to mine cryptocurrency without their consent, which can slow down the device and increase energy consumption.
- To protect against cybercrime targeting computers and mobiles, individuals and organizations should implement robust cybersecurity measures, regularly update software, use strong passwords, be cautious when clicking on links or downloading files and stay informed about the latest cyber threats and best practices. Fig. 2.4 Cyber Crime Targeting Computers and Mobiles illustrates attacks such as malware infection, unauthorized access, data breaches and device misuse aimed at compromising digital systems and mobile devices.

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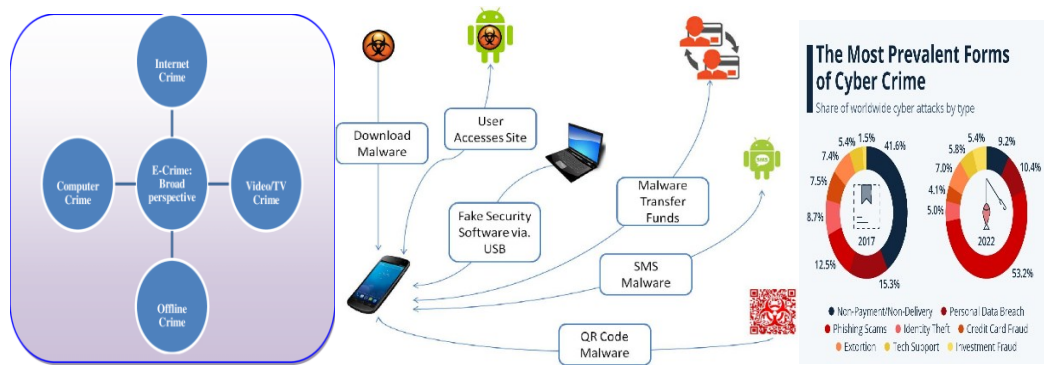


Fig. 2.4 Cyber crime targeting computers and mobiles

2.2.2 Cyber crime against women and children

Cyber crimes against women and children refer to offenses committed using computers, mobile devices and the internet that adversely affect their safety, dignity, privacy and psychological well-being. With the rapid growth of digital communication and social networking platforms, women and children have become vulnerable to misuse of technology for harassment, exploitation, intimidation, defamation and abuse. Such crimes often involve violation of personal privacy, unauthorized circulation of sensitive information or images and manipulation through online communication channels. Children are particularly at risk due to their lack of awareness and maturity, making them easy targets for online exploitation, while women frequently face gender-based abuse in cyberspace. These cyber offenses not only cause emotional and social harm but may also lead to long-term psychological trauma.

□ Cybercrimes against women and children are particularly concerning because they often involve harassment, exploitation or abuse of vulnerable individuals. Here are some common types of cybercrimes targeted at women and children:

1. **Cyberbullying:** Both women and children can be victims of cyberbullying, which includes online harassment, threats and intimidation. Perpetrators may use social media, messaging apps or other digital platforms to target their victims.
2. **Online Harassment:** This includes sending unsolicited, offensive or threatening messages, images or videos to women or children. It can be a form of cyberbullying and may have severe emotional and psychological effects.

3. **Revenge Porn:** Perpetrators may share explicit or intimate images or videos of women without their consent, often as an act of revenge. This is a violation of privacy and can cause significant harm to victims.
 4. **Sexting Exploitation:** In cases involving children, sexting can lead to exploitation when someone coerces or blackmails minors into sharing explicit images or videos. This can have legal and psychological consequences for the child involved.
 5. **Online Grooming:** Predators may use online platforms to groom children for sexual exploitation. They build trust with the child and gradually manipulate them into sharing personal information or engaging in inappropriate activities.
 6. **Child Pornography:** The distribution, possession, or creation of child pornography is illegal and exploits children. Criminals often use the internet to share such material.
 7. **Online Trafficking:** Human traffickers may use the internet to lure and exploit women and children, including for purposes of forced labor or sexual exploitation. Online platforms can be used to recruit victims.
 8. **Cyberstalking:** This involves persistent and unwanted online attention, often leading to fear or emotional distress. Women and children can be targeted by cyberstalkers who may threaten or harass them through digital means.
 9. **Financial Fraud:** Women can also be victims of financial fraud, including online scams targeting personal finances or online dating scams where perpetrators exploit emotional connections for financial gain.
 10. **Privacy Violations:** Privacy breaches can occur when personal information or photographs are shared without consent, affecting both women and children. This can lead to identity theft or other forms of cybercrime.
- To combat cybercrimes against women and children, various organizations and governments have implemented laws and initiatives aimed at raising awareness, providing support to victims, and prosecuting offenders.

Fig. 2.5 Online Violence against Children illustrates harmful activities such as cyber grooming, exploitation, bullying and exposure to inappropriate content that threaten the safety and well-being of minors in digital environments.



Fig. 2.5 Online Violence against Children

2.2.3 Financial frauds

Financial frauds in the cyber domain refer to illegal activities carried out using computers, networks and the internet with the intention of unlawfully gaining monetary benefits or causing financial loss to individuals, organizations or financial institutions. The advancement of electronic banking, online trading, and digital payment systems has increased the scope of such frauds, as offenders exploit technological vulnerabilities and user ignorance. These crimes involve unauthorized access to financial information, manipulation of electronic transactions, misuse of digital credentials, and deception through online communication. Financial frauds not only result in direct economic loss but also undermine trust in digital financial systems and electronic commerce.

- Financial frauds can have devastating consequences for individuals and the economy as a whole. While digital payments have made life convenient and easy In India, they have also made us prone to all kinds of financial frauds.

A. Ponzi Schemes: A Mirage of False Promises

- Ponzi schemes lure investors with promises of unusually high returns in a short period. The fraudsters use funds from new investors to pay off earlier investors, creating a false illusion of profitability.
- One infamous example is the **Saradha chit fund scam**, where millions of investors lost their hard-earned money. The group, consisting of over 200 private companies, falsely portrayed its collective investment schemes as chit funds.
- With an estimated collection of ₹200 to 300 billion (US\$4–6 billion), the scheme managed to attract deposits from more than 1.7 million individuals before its eventual downfall.

B. Identity fraud

- Identity fraud is common on Internet. Criminals have a few options when it comes to stealing your sensitive information.
- They might target you with a phishing attack where they email, call, or text pretending to be from your bank. Or, they could target you with a cyber attack to get you to install malware on your devices that steals your logins and passwords.
- How do you know you're being targeted?
 - Unfamiliar transactions on your credit card.
 - Strange charges on your bank statements.
 - New credit cards or loans in your name.
 - Missing or error-filled tax returns.
 - Calls from debt collectors about purchases you didn't make.
 - A drop in credit score.
 - Bounced checks.

C. Fraudulent charities

- Scammers use philanthropy as fraud, too. Charity fraud entails creating a fake charity and collecting “donations” that disappear along with the thief
- How does charity fraud happen?

- Scammers create fake charities — like military veteran charities — that sound like ones you know and trust. These scams are especially common during natural disasters or international news events.
- What are the warning signs?
- Claiming that you're a previous donor when you know you've never sent them money.
- Only accepting donations through cash, cryptocurrency, gift cards, or wire transfers

D. Credit card fraud

- There are several ways that criminals can steal your credit card information. They could steal your physical card, trick you into entering information on a phishing website or email, buy your details on the Dark Web, or use any number of other credit card scams.
- Hackers can also create a clone of your physical card using just your credit card numbers.
- What are the warning signs?
 - Suspicious transactions on your credit card or bank statement.
 - Small unfamiliar charges on your account. (Fraudsters use a scam called carding to validate your credit card before making large purchases.)
 - Fraud alerts from your bank, credit card issuer, or credit monitoring service.

E. Stock Market Manipulation

- Stock market manipulation includes activities like price rigging, spreading false information, insider trading, and pump-and-dump schemes. Fraudsters manipulate stock prices, deceiving investors and causing significant financial losses.
- The Satyam Computer Services scandal is a prime example, where the company's promoters manipulated financial statements to inflate stock prices.
 - **Bank Frauds**

- Bank frauds encompass various fraudulent activities, including loan frauds, cheque frauds, forged documents, and unauthorized transactions. These frauds result in substantial financial losses for banks and individuals.
- One notable case is the Nirav Modi-PNB scam, where fraudulent Letters of Undertaking were issued, causing a massive loss to Punjab National Bank.

F. How to protect yourself against financial frauds

1. Protect your personal information
2. Monitor financial activities
3. Be cautious online
4. Use strong passwords and enable two-factor authentication
5. Stay informed about scams
6. Keep your devices secure
7. Exercise caution with public Wi-Fi
8. Verify before sharing information

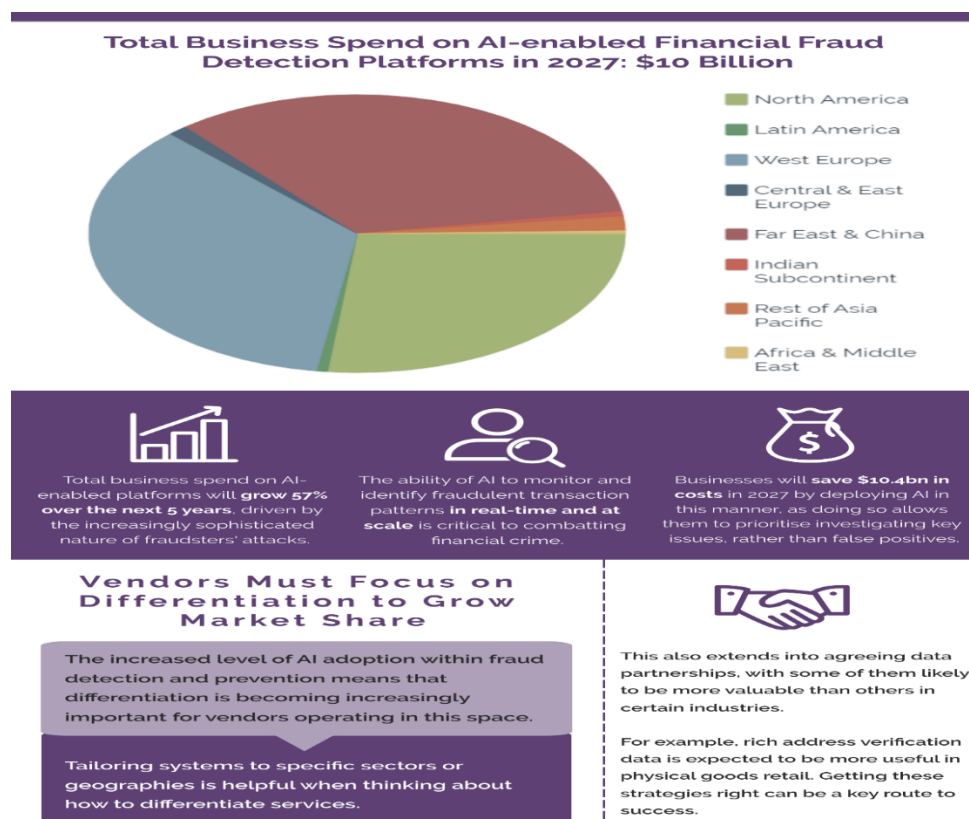


Fig. 2.6 AI in Financial Fraud Detection – 2022-2027 Market Summary

Fig. 2.6 AI in Financial Fraud Detection – 2022–2027 Market Summary highlights the growing adoption of artificial intelligence to identify, prevent and mitigate financial fraud through advanced data analytics and real-time monitoring.

2.2.4 Social Engineering Attacks

Social engineering attacks manipulate human psychology to gain unauthorized access to information, systems or resources.

A. Social Engineering

- It is the “technique to influence” & “persuasion to deceive” people to obtain the information.
- It exploits the fact that people are the weak link in security.
- Social engineers build the trust with the victim/person to gain the unauthorized information/access
- Their goal is to fool someone into providing valuable information.
- Example: The attacker (social engineer) calling a user & pretending to be a tech support person & ask questions about the confidential files, passwords, etc.

B. Classification of Social Engineering

1. Human based Social Engineering:

- It refers to person to person interaction to get the unauthorized information.
- The following are its different types.
 - i. Impersonating an employee or valid user: The attacker impersonates/poses as an employee of the same organization to take the advantage from the people who are helpful.
 - ii. Posing as important user: The attacker pretends to be a CEO/Manager who intimidates lower level employee in order to gain access to the system.
 - iii. Using a third person: The attacker pretends to have permission from an authorized source/person (who cannot be contacted for verification) to use a system.
 - iv. Calling technical support: Attacker calls help desk or tech support personnel to obtain the information since they are trained to help users.

- v. Shoulder surfing: It involves gathering information (usernames, passwords, etc) by watching over a person's shoulder while he/she logs into the system.
- vi. Dumpster diving (Scavenging/Binning): It involves looking in the trash/dustbin for information written on pieces of paper, computer print outs, etc.

Fig. 2.7 Anatomy of Social Engineering Attacks illustrates the stages of deception, trust manipulation, information extraction, and exploitation used by attackers to trick individuals into revealing sensitive data. Fig. 2.8 The Social Engineering Threat Landscape illustrates the range of human-centric attack methods, including phishing, pretexting, baiting and impersonation, used to exploit trust and psychological vulnerabilities.



Fig. 2.7 Anatomy of Social Engineering Attacks



Fig. 2.8 The Social Engineering Threat Landscape

2. Computer based Social Engineering

- It refers to the attempts made to get the unauthorized information by using computer/software/internet.
- The following are its different types.
 - i. **Fake emails:** It involves the attacker sending fake emails (pretending as a legitimate email) to a number of users in order to make the users to reveal their sensitive information such as usernames, passwords, credit card details, etc. It is also called as Phishing.
 - ii. **Email attachments:** It involves sending malicious codes to victim's system in the form of an email attachment. The virus, worms, etc which will be present in the email attachment will be automatically executed if the victim opens the attachment.
 - iii. **Pop-up windows:** They are used similar to email attachments but they encourage the victim to click on special offers or free stuffs so that the malicious code can be installed to the system.

Fig. 2.9 Common Types of Phishing Attacks illustrates deceptive techniques such as email phishing, spear phishing, smishing, vishing and clone phishing used to steal sensitive information.

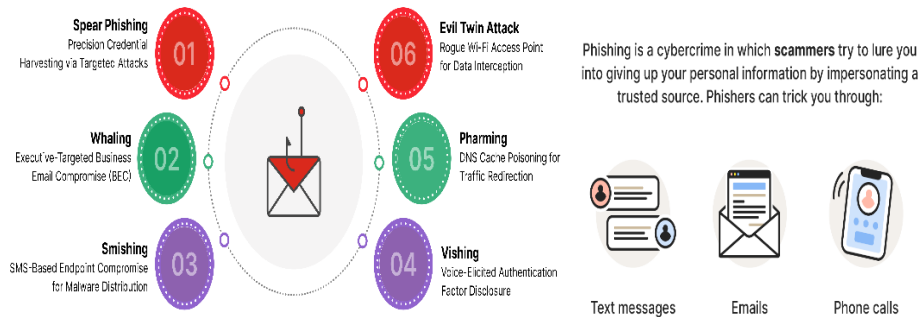


Fig. 2.9 Common Types of Phishing Attacks

3. Effects of Social Engineering:

- ❑ Loss/altering of medical & healthcare information, corporate financial data, electronic funds transfers, etc.
- ❑ Loss of customers
- ❑ Loss of funds
- ❑ Loss of trust
- ❑ Collapse of the organization

4. Counter measures (Security) against Social Engineering:

- ❑ Providing training/awareness to the potential victims at regular intervals about the attacks
- ❑ Creating awareness on how attackers gain the trust of the victims
- ❑ Strict policies about service desk staff, not to ask for personal/sensitive information
- ❑ Educate potential victims to recognize social engineering attempt

2.2.5 Malware and Ransomware attacks

Malware attacks refer to the use of malicious software such as viruses, worms, Trojans, spyware and bots that are intentionally designed to damage systems, disrupt operations or steal sensitive personal, financial or organizational data without the user's knowledge. These attacks commonly spread through infected email attachments, malicious websites, fake

software downloads or removable storage devices. **Ransomware** is a specific and severe type of malware that encrypts files or locks the entire system and then demands a ransom payment from the victim to restore access. Ransomware attacks usually enter a system through phishing emails, unsafe links or remote desktop vulnerabilities and can cause serious consequences such as data loss, financial damage, operational downtime and loss of trust, making them one of the most dangerous cyber threats today.

2.2.5.1 Malware Attacks

- Malware attacks are any type of malicious software designed to cause harm or damage to a computer, server, client or computer network and/or infrastructure without end-user knowledge
- Cyber attackers create, use and sell malware for many different reasons, but it is most frequently used to steal personal, financial or business information.

Fig. 2.10 Types of Malware illustrates malicious software categories such as viruses, worms, trojans, spyware, ransomware and adware that compromise system security.

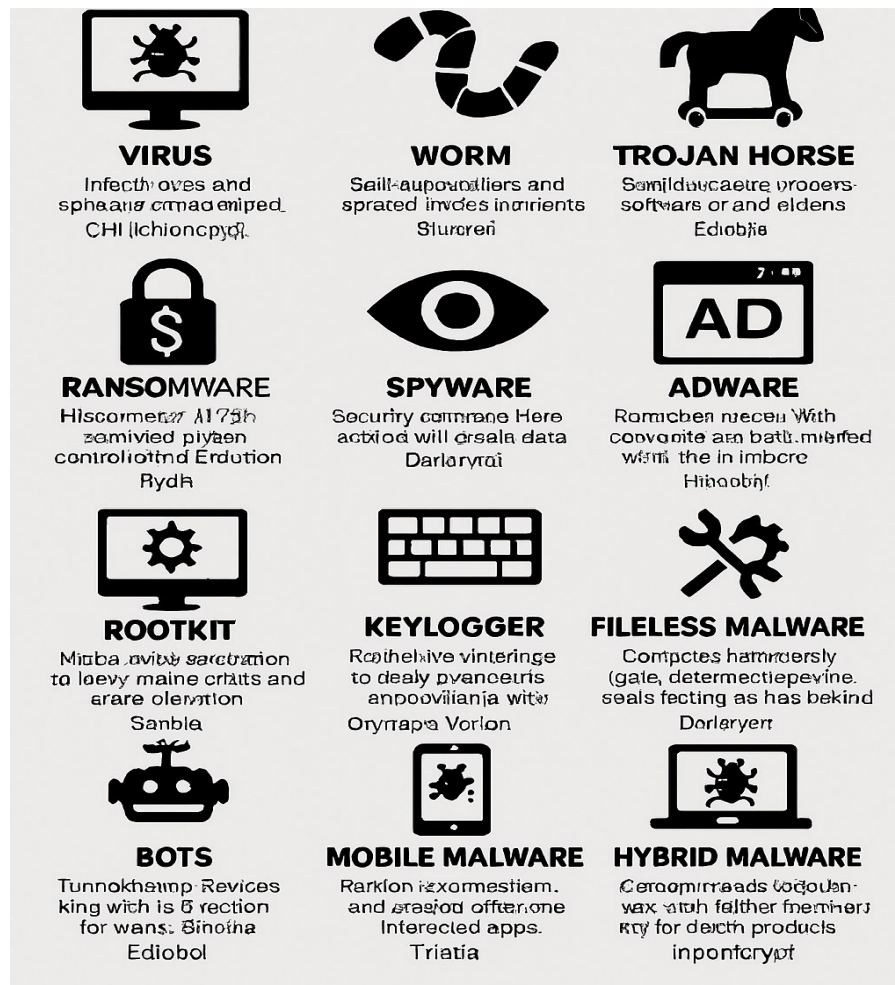


Fig. 2.10 Types of Malware

A. Types of Malware

1. **Adware:** Display ads (sometimes malicious ads) to users as they work on their computers or browse the web.
2. **Viruses:** A virus infects a computer and performs a variety of payloads. It may corrupt files, destroy operating systems, delete or move files, or deliver a payload at a specific date.
3. **Worms:** A worm is a self-replicating virus, but instead of affecting local files, a worm spreads to other systems and exhausts resources.
4. **Trojans:** A Trojan is named after the Greek war strategy of using a Trojan horse to enter the city of Troy. The malware masquerades as a harmless program, but it runs

in the background stealing data, allowing remote control of the system, or waiting for a command from an attacker to deliver a payload.

5. **Bots:** Infected computers can become a part of a botnet used to launch a distributed denial- of-service by sending extensive traffic to a specific host.
6. **Keyloggers:** Capture keystrokes as users type in URLs, credentials, and personal information and send it to an attacker.
7. **RAT:** “Remote access tools” enable attackers to access and control the targeted device remotely.
8. **Downloaders:** Download other malware to install locally. The type of malware depends on the attacker’s motives.
9. **POS:** Compromise a point-of-sale (PoS) device to steal credit card numbers, debit card and PINs, transaction history, and contact information.

B. How do I know I’ve been infected with malware?

- ☐ The most common signs that your computer has been compromised by malware are:
- ☐ Slow computer performance
- ☐ Browser redirects, or when your web browser takes you to sites you did not intend to visit
- ☐ Infection warnings, frequently accompanied by solicitations to buy something to fix them
- ☐ Problems shutting down or starting up your computer
- ☐ Frequent pop-up ads

C. How can I protect myself from malware?

1. Protect your devices

- ☐ Keep your operating system and applications updated. Cybercriminals look for vulnerabilities in old or outdated software, so make sure you install updates as soon as they become available.
- ☐ Never click on a link in a popup. Simply close the message by clicking on “X” in the upper corner and navigate away from the site that generated it.

- Limit the number of apps on your devices. Only install apps you think you need and will use regularly. And if you no longer use an app, uninstall it.

2. Be careful online

- Avoid clicking on unknown links. Whether it comes via email, a social networking site or a text message, if a link seems unfamiliar, keep away from it.
- Be selective about which sites you visit. Do your best to only use known and trusted sites,
- Beware of emails requesting personal information. If an email appears to come from your bank and instructs you to click a link and reset your password or access your account, don't click it. Go directly to your online banking site and log in there.
- Avoid risky websites, such as those offering free screensavers.

3. Perform regular checks

- If you are concerned that your device may be infected, run a scan using the security software you have installed on your device.
- Check your bank accounts and credit reports regularly.

2.2.5.2 Ransomware Attack

- A ransomware attack is a malware that encrypts personal information and documents while demanding a ransom amount to decrypt them.
- Once the files are encrypted or locked behind a password, a text file is available to the victim, explaining how to make the ransom payment and unlock the files for it.

A. How Does a Ransomware Attack Work?

- The spread of ransomware mostly starts with phishing attacks. A ransomware attack gains access to a victim's device through infected emails, messages, and malicious sites and encrypts the data in that device.
- The ransomware uses simple asymmetric encryption algorithms, blocks a user's files, and makes them difficult to decrypt without knowing the key.
- Another way to breach a system with ransomware is by using the Remote Desktop Protocol or RDP access. It can access remotely a computer using this protocol,

allowing a hacker to install malicious software on the system with the owner, unaware of these developments.

- ❑ Ransomware adds instruction files describing the pay-for-decryption process, then uses those files to present a ransom note to the user.
- ❑ Ransomware usually terminates and destroys itself by leaving only the payment instruction files.

Fig. 2.11 Anatomy of a Crypto-Ransomware Attack illustrates the stages of infection, encryption of victim data, ransom demand and attempted extortion for decryption keys.

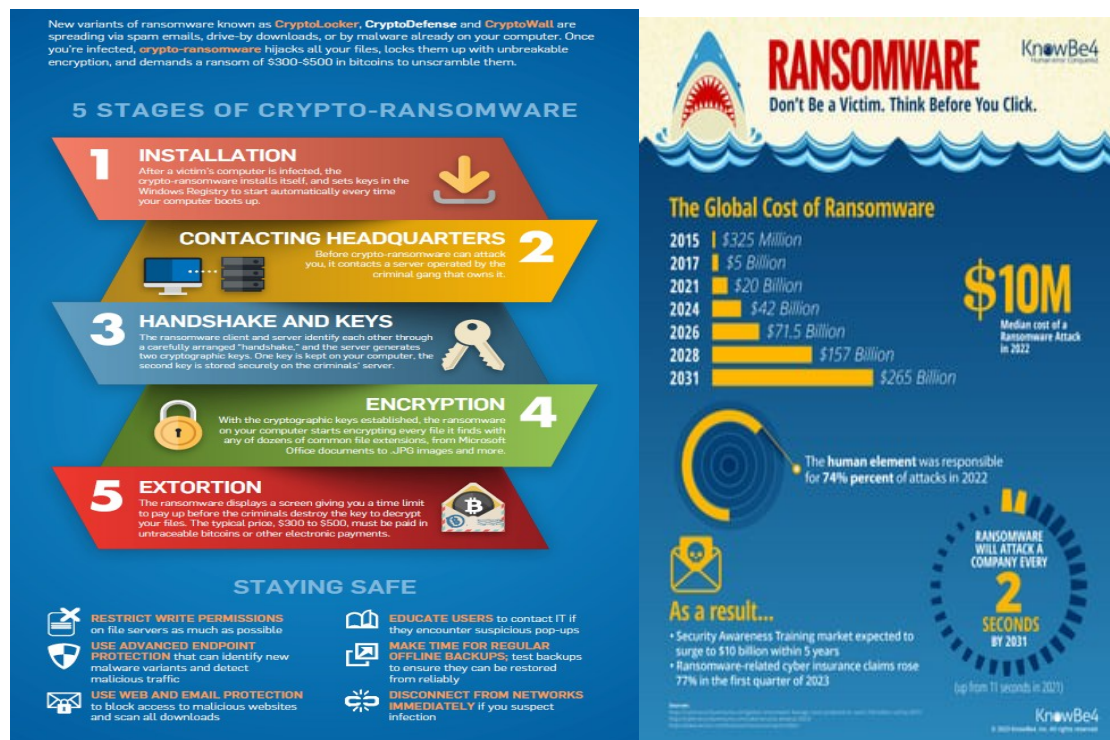


Fig. 2.11 Anatomy of a Crypto-Ransomware Attack

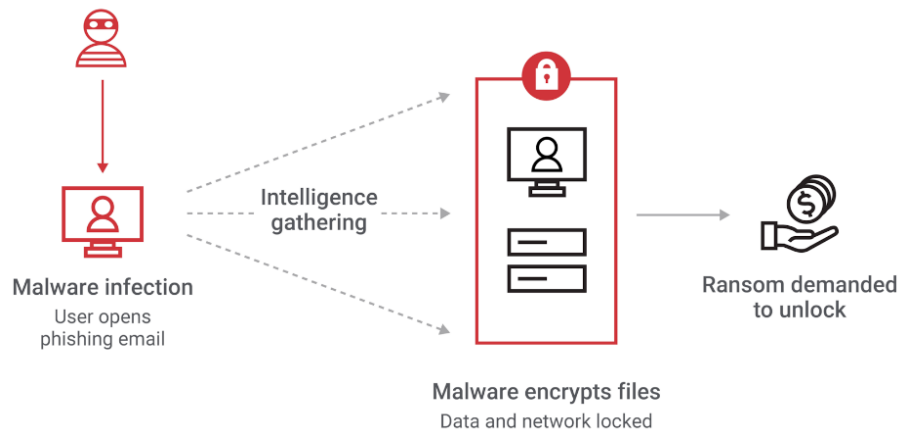


Fig. 2.12 How Ransomware Works

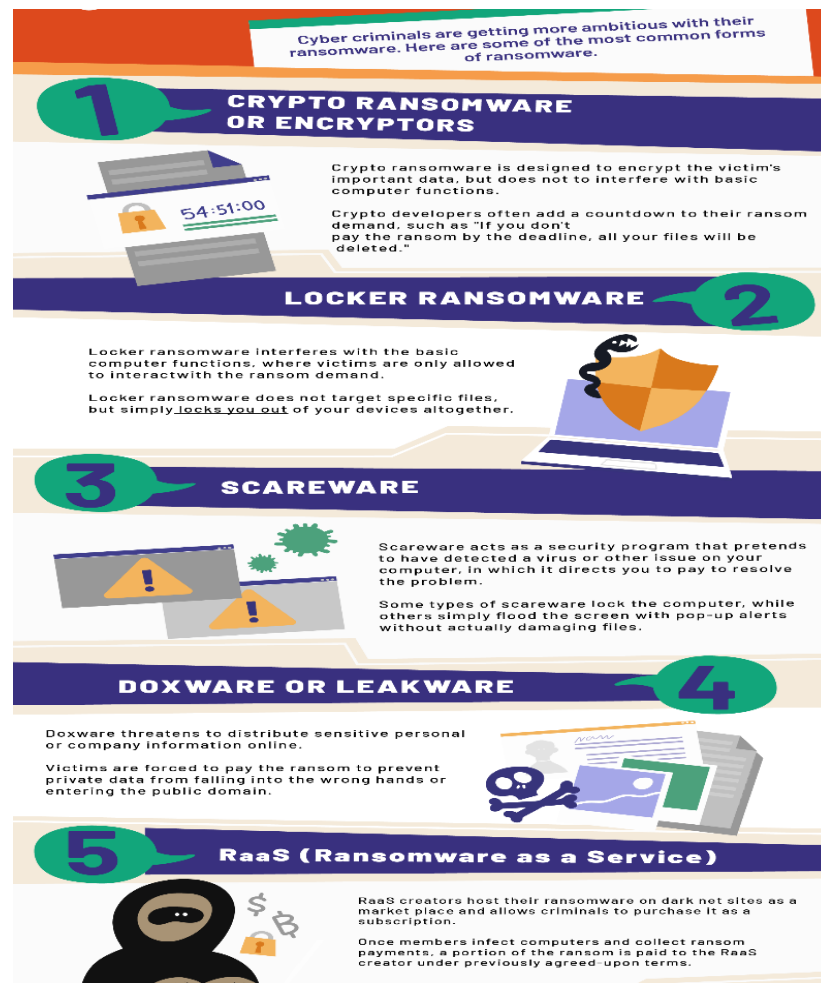


Fig. 2. 13 Different forms of Ransomware

Fig. 2.12 & 2.13 illustrates how ransomware works and Different Forms of Ransomware shows the major ransomware variants such as crypto-ransomware, locker ransomware, scareware and leakware that extort victims in different ways.

B. Types of Ransomware

1. Locker ransomware

- It is a type of malware that blocks standard computer functions from being accessed until the payment to the hackers is not complete.
- It shows a lock screen that doesn't allow the victim to use the computer for primary purposes.

2. Crypto ransomware

- This ransomware encrypts the local files and documents on the computers.
- Once the files are encrypted, finding the decryption key is impossible unless the ransomware variant is old and the keys are already available on the internet.

3. Scareware

- It is a fake software that claims to have detected a virus or other issue on your computer and directs you to pay to resolve the problem.
- Some scareware locks the computer, while others flood the screen with pop-up alerts without damaging files.

C. How to Prevent Ransomware Attacks?

- One must always have backups of their data. Cloud storage for backup is easy, but a physical backup in a hard drive is always recommended.
- Keeping the system updated with the latest security patches is always a good idea.
- Apart from system updates, one must always have reputed antivirus software installed.
- If a system is infected with ransomware already, there is a website, 'nomoreransom.org.' It has a collection of decryption tools for most well-known ransomware packages.

2.2.6 Zero day and Zero click attacks Zero day

- Software often has security vulnerabilities that hackers can exploit to cause havoc.
- The term "zero-day" refers to the fact that the vendor or developer has only just learned of the flaw – which means they have “zero days” to fix it.
- A zero-day attack takes place when hackers exploit the flaw before developers have a chance to address it.
- Zero-day attackers can steal data, corrupt files, take control of devices, install malware or spyware, and more.
- Typical targets for a zero-day exploit include:
 1. Government departments.
 2. Large enterprises.
 3. Individuals with access to valuable business data, such as intellectual property.
 4. Hardware devices, firmware and Internet of Things (IoT).

Fig.2.14 & Fig.2.15 Zero-Day Threat Lifecycle illustrates how undiscovered software vulnerabilities are identified and exploited by attackers, resulting in active cyber attacks until the flaw is disclosed and security patches are released.

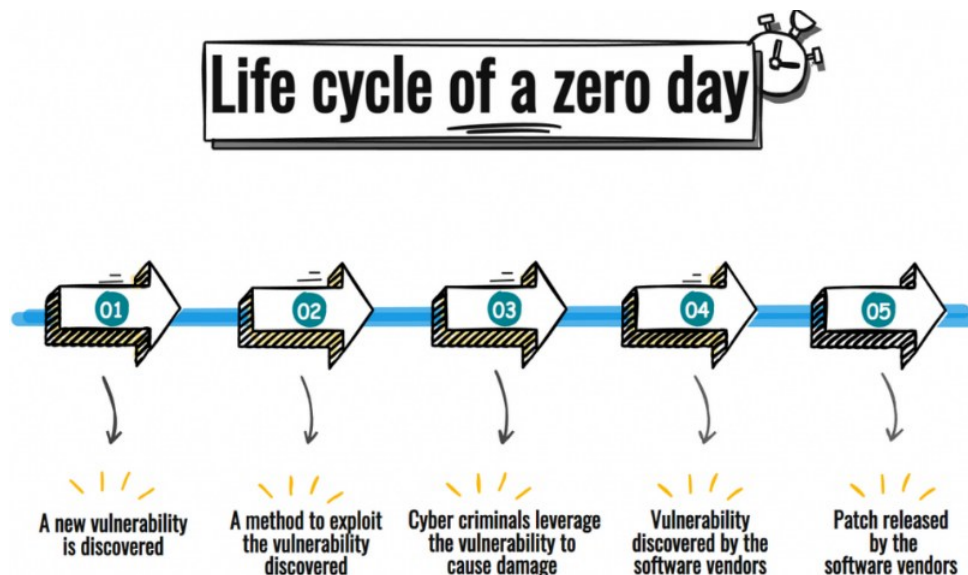


Fig. 2.14 Life cycle of a Zero day

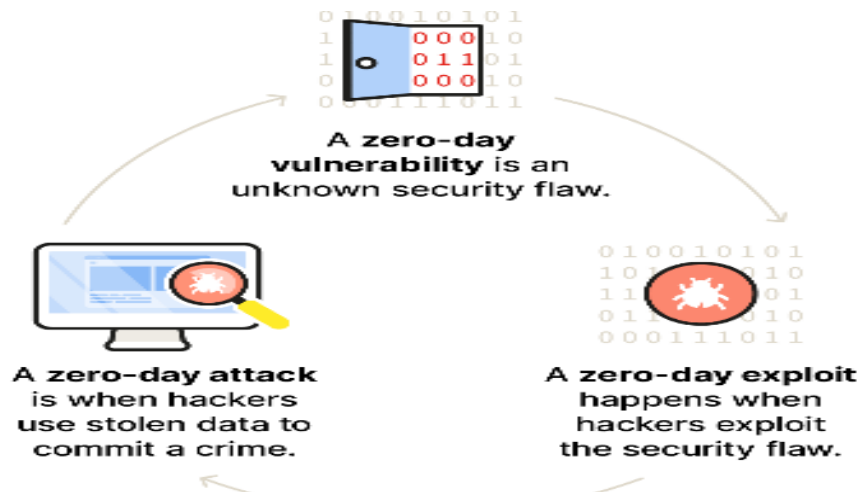


Fig. 2.15 Zero-Day threats: From Vulnerabilities to attacks

A. Recent Examples of Zero Day Attacks

- In December 2021, Amazon Web Services, Microsoft, Cisco, Google Cloud, and IBM were among the major tech players affected by the Log4j vulnerability in an open-source logging library.
- In 2021, Google's Chrome suffered a series of zero-day threats, causing Chrome to issue updates. The vulnerability stemmed from a bug in the V8 JavaScript engine used in the web browser.
- Zoom was targeted in 2020. Hackers were able to remotely access users' PCs if the video conferencing platform was running on an older version of Windows.
- Apple's iOS fell victim in 2020 to two sets of zero-day bugs that saw attackers compromising iPhones remotely.

B. How to protect yourself against zero-day attacks

1. **Keep all software and operating systems up to date.** This is because the vendors include security patches to cover newly identified vulnerabilities in new releases. Keeping up to date ensures you are more secure.
2. **Use only essential applications.** The more software you have, the more potential vulnerabilities you have. You can reduce the risk to your network by using only the applications you need.

3. **Use a firewall.** A firewall plays an essential role in protecting your system against zero-day threats. You can ensure maximum protection by configuring it to allow only necessary transactions.

C. Zero click

- zero-click attacks require no action from the victim – meaning that even the most advanced users can fall prey to serious cyber hacks and spyware tools.
- also called interaction-less or fully remote attacks.
- spying software relies on convincing the targeted person to click on a compromised link or file to install itself on their phone, tablet, or computer.
- However, with a zero-click attack, the software can be installed on a device without the victim clicking on any link. As a result, zero-click malware or no-click malware is much more dangerous.
- The target of a zero-click attack can be anything from a smartphone to a desktop computer and even an IoT device

Fig. 2.16 & Fig. 2.17 Zero-Click Attacks illustrate how attackers exploit hidden software vulnerabilities to automatically compromise devices without any user interaction, enabling silent execution of malicious code and unauthorized access.



Fig. 2.16 Zero-Click Attack Process in Steps

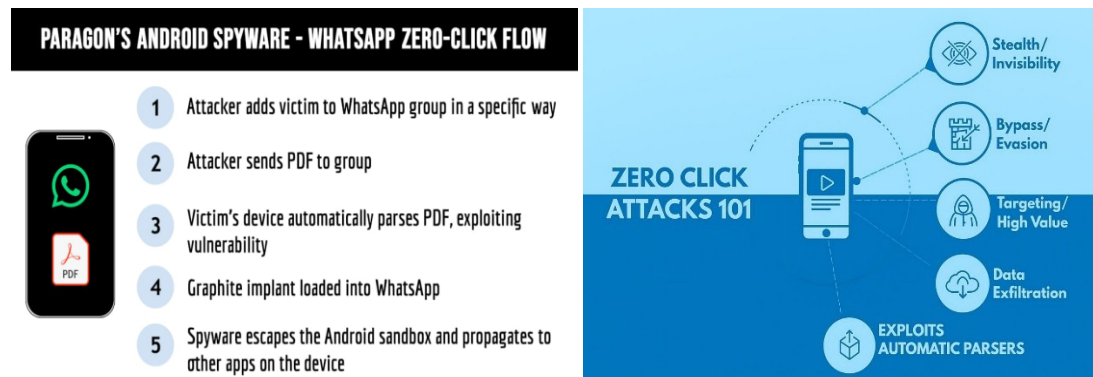


Fig. 2.17 Zero Click Attacks

D. Examples of Zero-Click Attacks

1. **Apple zero-click, forced entry, 2021:** In 2021, a Bahraini human rights activist had their iPhone hacked by powerful spyware sold to nation-states.
2. **WhatsApp breach, 2019:** This infamous breach was triggered by a missed call, which exploited a flaw in the source code framework of WhatsApp.

E. How to protect yourself from zero-click exploits

- ❑ Keep your operating system, firmware, and apps on all your devices up to date as prompted.
- ❑ Only download apps from official stores.
- ❑ Delete any apps you no longer use.
- ❑ Use your device password protection.
- ❑ Use strong authentication to access accounts, especially critical networks.
- ❑ Use strong passwords – i.e., long and unique passwords.

2.3 Modus Operandi of Cyber Criminals

The modus operandi of cyber criminals refers to the systematic methods and techniques used to carry out cyber crimes. Typically, cyber criminals begin by gathering information about the target through social engineering, phishing, or reconnaissance. They then exploit system vulnerabilities, weak passwords or user trust to gain unauthorized access. Once access is obtained, malicious activities such as data theft, malware installation, financial fraud, or system disruption are carried out. Finally, attackers attempt to cover their tracks by

deleting logs, using anonymization tools or routing attacks through multiple systems to avoid detection, making investigation and tracing difficult.

In general, modus operandi is the method acquired by any criminal for the successful commission of a crime. At a minimum, every Modus Operandi will contain three basic elements namely:

1. Ensure success of the crime
2. Protect identity
3. Facilitate effective escape

2.3.1 Common forms of modus operandi

1. Sending Annoying Messages

- Annoying, Insulting, Misleading, Defaming messages are often sent using mobile phones in bulk. Hence the actual source could not be fixed.
- Such messages are often a cause of misperception among people of different race, culture and tradition many a times often resulting in fights or riots.
- Unaware and innocent people often fall in traps of cyber criminals for SMS of lottery, Emails of prize money, false promise of jobs, and false mail for admission in reputed colleges.
- Multimedia messages often defaming the identity of a person are distributed among small groups using mobile phones.
- Pornography, Obscene messages and cyber bullying are becoming very common and very popular, for e.g. Delhi MMS Scandal.
- Obscene videos are often captured in remote places unknowingly of the victim for future exploitation.

2. Making Offensive Calls

- Offenders can also harass others by making offensive calls to them and annoying them.
- Many a time anonymous calls are used by the criminals as an effective tool in making extortion or threatening call. Females are often harassed by stalkers by this means of communication.

- Landlines having no Caller Ids pose a problem for the quick analysis of an incoming call, which is an undue advantage to the cyber stalkers, cyber bullies, etc.
- Calls can be made by spoofing the mobile number using various sites. Such calls are intended to hide the actual location of the caller and any fake or annoying calls are made. Such calls are often used for terrorist activity and for trafficking illegal goods or for any ransom or blackmailing purposes.
- Cyber Criminals operating from overseas and indulged in forgery are hard to trace without the co-operation of international agencies.

2.4 Reporting of cyber crimes

Reporting of cyber crimes is the process of informing authorized agencies about incidents such as online fraud, hacking, identity theft, cyber bullying, or data breaches so that timely investigation and legal action can be taken. Victims should immediately preserve evidence like emails, messages, screenshots, transaction details, and log files related to the incident. In India, cyber crimes can be reported through the National Cyber Crime Reporting Portal or at the nearest cyber crime police station. Prompt reporting helps prevent further damage, assists law enforcement in tracking criminals, and creates awareness about emerging cyber threats, thereby strengthening overall cyber security.

Reporting cybercrimes is essential to combat online threats and hold perpetrators accountable. Here are the steps you can take to report cybercrimes:

- 1. Contact Your Local Law Enforcement:** If you believe you are a victim of a cybercrime, you should report it to your local police department or law enforcement agency. They can investigate the incident and take appropriate action.
- 2. Report to a National Cybersecurity Agency:** In many countries, there are dedicated agencies responsible for handling cybercrimes. In the United States, for example, you can report cybercrimes to the Federal Bureau of Investigation (FBI) through their Internet Crime Complaint Center (IC3). Check if your country has a similar agency and report the incident to them.
- 3. Report to the Appropriate Online Platforms:** If the cybercrime occurred on a specific online platform, such as a social media site, email service, or e-commerce

website, report the incident to that platform. They may have mechanisms in place to address various online abuses and can take action against the responsible parties.

- 4. Report to Anti-Fraud Organizations:** There are organizations like the Anti-Phishing Working Group (APWG) and the Anti-Malware Testing Standards Organization (AMTSO) that collect information about cyber threats and work with law enforcement. Reporting incidents to these organizations can help in identifying trends and patterns.
- 5. Report to Financial Institutions:** If the cybercrime involves financial fraud, contact your bank or financial institution immediately. They can help you secure your accounts and investigate any unauthorized transactions.
- 6. Report to Internet Service Providers (ISPs):** If you have evidence of cybercrimes, such as hacking or distribution of illegal content, involving an IP address, contact the relevant Internet Service Provider (ISP). They may take action against the offender or provide assistance to law enforcement.
- 7. Document the Incident:** Make sure to document all evidence related to the cybercrime, including emails, messages, screenshots, IP addresses, and any other relevant information. This documentation can be crucial for investigations.
- 8. Use Online Reporting Portals:** Many countries and regions have online reporting portals where you can report cybercrimes. These portals may be managed by government agencies or law enforcement. Check if your region offers such a service.
- 9. Consider Legal Advice:** In some cases, it may be necessary to seek legal advice or consult with a cybersecurity expert to understand the best course of action and to help with the investigation.
- 10. Protect Yourself:** While reporting the cybercrime, take steps to secure your online presence, change passwords, update security settings, and install or update security software to prevent further incidents.

Remember that reporting cybercrimes is essential for both your own protection and the collective effort to combat online threats. The information you provide can help authorities take action and prevent future cybercrimes.

2.5 Remedial and mitigation measures

Remedial and mitigation measures are actions taken to reduce the impact of cyber incidents and prevent future attacks. These measures include immediately isolating affected systems, changing compromised passwords, restoring data from secure backups and removing malicious software using updated antivirus tools. Regular software updates, strong authentication mechanisms, user awareness training, and strict security policies help mitigate risks. Additionally, monitoring systems continuously, maintaining incident response plans, and promptly reporting cyber incidents to authorities play a crucial role in minimizing damage and improving organizational cyber resilience. Fig. 18 illustrates the legal framework governing cyber crime in India, highlighting the role of the Information Technology Act, 2000, its amendments, and supporting provisions of the Indian Penal Code in regulating cyber offences and ensuring digital security.

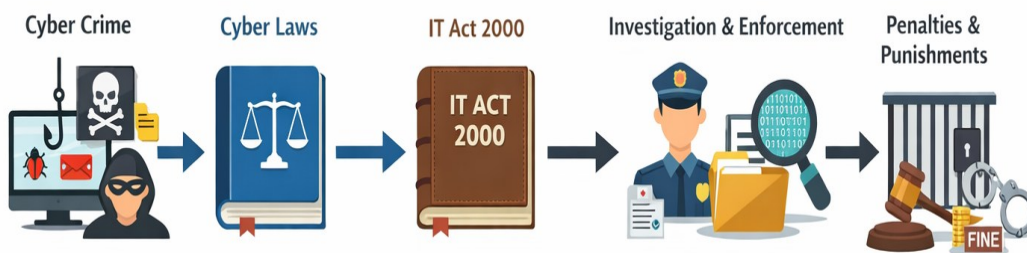


Fig. 2.18 Legal framework governing cyber crime in India,

2.5.1 Remedial Measures:

- 1. Incident Response:** In the event of a cyber crime, organizations should have an incident response plan in place to quickly identify, contain, and mitigate the impact of the attack. This includes isolating affected systems, restoring backups, and applying patches or security updates.
- 2. Forensic Investigation:** Engaging professional forensic investigators can help identify the source and extent of the cyber crime, gather evidence, and aid in **Legal perspective of** proceedings.

- 3. Data Recovery:** If data is compromised or encrypted due to a cyber attack, organizations should have backups in place to restore affected systems and minimize data loss.

2.5.2 Mitigation Measures:

- 1. Strong Security Practices:** Implement robust security measures, such as firewalls, antivirus software, and intrusion detection and prevention systems, to protect against cyber threats.
- 2. Regular Updates and Patching:** Keep software, operating systems, and firmware up to date with the latest security patches to mitigate vulnerabilities that cyber criminals may exploit.
- 3. Employee Education:** Provide cybersecurity awareness and training programs to employees to educate them about common cyber threats, phishing techniques, and safe online practices.
- 4. Multi-factor Authentication (MFA):** Implement MFA wherever possible to add an extra layer of security, making it harder for cyber criminals to gain unauthorized access to accounts or systems.
- 5. Data Encryption:** Encrypt sensitive data, both in transit and at rest, to ensure that even if it is intercepted or stolen, it remains unreadable and unusable for unauthorized individuals.
- 6. Regular Security Audits:** Conduct regular security audits and vulnerability assessments to identify and address any weaknesses or potential entry points for cyber criminals.

2.6 Legal perspective of cyber crime

From a legal perspective, cyber crime refers to unlawful acts committed using computers, networks or the internet, which are governed by specific cyber laws to ensure security, justice, and accountability in the digital environment. With the rapid growth of information technology and widespread internet usage, societies across the world have become increasingly digitally sophisticated, and correspondingly, criminal activities have also

evolved in complexity and scale. As a result, the necessity for cyber laws and their effective application has gained significant momentum.

All legal issues related to internet-based offences are addressed through cyber laws, which provide a structured framework to recognize risks arising from the use of computers, computer networks, and electronic communication systems. Cyber law gives legal recognition to activities conducted in cyberspace and defines the responsibilities and liabilities of users and service providers. The legal framework aims to protect individuals, organizations, and government institutions by clearly defining cyber offences, establishing investigation and enforcement mechanisms, and promoting lawful behavior in cyberspace.

Cyber law broadly encompasses laws relating to:

- Cyber crimes
- Electronic and digital signatures
- Intellectual property
- Data protection and privacy

Thus, the legal perspective of cyber crime focuses on regulating digital conduct, preventing misuse of technology, safeguarding sensitive information and ensuring a secure and trustworthy digital ecosystem.

Legal perspective of cybercrime in India

- In India, cybercrime is primarily governed by the Information Technology Act, 2000 (IT Act). This law was established to address various cyber offenses and provide a legal framework for electronic transactions, digital signatures, and data protection.
- The purpose of the Indian IT Act(ITA) was to amend the Indian Penal Code(IPC).

2.6.1 Information Technology Act, 2000 and Its Amendments

In India, cyber crime is primarily governed by the Information Technology Act, 2000 (IT Act), which serves as the principal legislation for addressing offences committed in the digital domain. The Act was enacted to provide legal recognition to electronic records and digital signatures, facilitate secure electronic transactions, and establish a comprehensive legal framework for cyber offence regulation. One of the important objectives of the Indian

IT Act (ITA) was to supplement and strengthen existing legal provisions, including those of the Indian Penal Code (IPC), to effectively deal with technology-related crimes.

The IT Act addresses a wide range of cyber offences such as unauthorized access to computer systems, damage to data or networks, hacking, identity theft, cyber terrorism, online fraud, breach of confidentiality and publication of obscene electronic content. Various sections of the Act prescribe penalties in the form of fines, compensation, and imprisonment, thereby ensuring deterrence and legal enforcement. The Act also assigns responsibilities to authorities and intermediaries to ensure compliance and cyber security.

2.6.1.1 Key Provisions under the Indian IT Act, 2000

The major provisions of the IT Act that deal with cyber offences and penalties are summarized below:

- **Section 43 – Penalty for damage to computer system, etc :** Provides compensation up to ₹1 Crore for damage to computer systems, networks, or data.
- **Section 66 – Hacking with computer system:** Deals with hacking committed with intent or knowledge, punishable with a fine up to ₹2 lakhs and/or imprisonment up to 3 years.
- **Section 67 – Publishing obscene information in electronic form:** Addresses publication or transmission of obscene electronic material, with punishment including fine and imprisonment, and enhanced penalties for repeat offences.
- **Section 68 – Power of Controller to give directions:** Penalizes non-compliance with lawful directions issued by the Controller, with fines or imprisonment.
- **Section 70 – Protected system:** Protects critical computer systems from unauthorized access, punishable with imprisonment up to 10 years.
- **Section 72 – Penalty for breach of confidentiality and privacy.** Deals with unauthorized disclosure of confidential information, punishable with fine or imprisonment.
- **Section 73 – Publishing false digital signature certificates** Addresses publication of false digital signature certificates with fraudulent intent.

- **Section 74 – Publication for fraudulent purpose** Penalizes fraudulent use of digital signature certificates.

2.7 Cyber crime and offences

Cybercrime encompasses various illegal activities conducted through digital means, often targeting individuals, organizations, or systems. Here are some common cybercrimes and offenses:

1. **Hacking:** Unauthorized access to computer systems, networks or devices to manipulate, steal data or disrupt operations.
2. **Identity Theft:** Stealing personal information (such as Social Security numbers, credit card details) to impersonate someone else, commit fraud or gain access to financial resources.
3. **Phishing and Spoofing:** Sending deceptive emails or creating fake websites to trick individuals into revealing sensitive information (passwords, financial data) or downloading malware.
4. **Cyberbullying:** Harassment, threats, or intimidation using digital platforms, often directed at individuals, which can have serious emotional and psychological effects.
5. **Online Fraud:** Illegitimate schemes to deceive individuals or entities for financial gain, including investment scams, online shopping fraud and auction fraud.
6. **Distributed Denial of Service (DDoS) Attacks:** Overloading servers or networks with excessive traffic to disrupt access, making websites or services unavailable to users.
7. **Cyber Espionage:** Unauthorized access to confidential information or intellectual property of governments, organizations or individuals, often carried out by other governments or corporate entities.
8. **Child Exploitation and Pornography:** Using digital means to produce, distribute, or possess child pornography or engage in illegal activities involving minors.

9. Ransomware Attacks: Malicious software that encrypts files or systems, demanding payment (usually in cryptocurrency) for decryption or to avoid data exposure.

10. Cyberstalking: Persistent harassment or monitoring of an individual online, causing fear or emotional distress.

2.7.1 Organizations dealing with Cybercrime and Cyber security in India,

In India, several organizations are involved in dealing with cybercrime and cybersecurity at various levels, including law enforcement, regulatory bodies and agencies focused on awareness and prevention.

Some prominent ones include:

- 1. National Cyber Security Coordinator (NCSC):** The NCSC operates under the Prime Minister's Office and is responsible for coordinating all cybersecurity initiatives in the country.
- 2. Computer Emergency Response Team-India (CERT-In):** CERT-In is the national nodal agency under the Ministry of Electronics and Information Technology that deals with cybersecurity incidents, response and related issues.
- 3. National Critical Information Infrastructure Protection Centre (NCIIPC):** NCIIPC is responsible for protecting critical information infrastructure in the country and formulating policies and guidelines for securing these assets.
- 4. State Police Cyber Cells:** Many states have established specialized cyber cells within their police departments to investigate and handle cybercrimes at the state level.
- 5. National Investigation Agency (NIA):** NIA deals with investigating and prosecuting offenses affecting the sovereignty, security and integrity of India, including cybercrimes with national implications.
- 6. Cyber Appellate Tribunal (CAT):** It hears appeals against any order passed by CERT-In or the Adjudicating Officer under the Information Technology Act, 2000.
- 7. Banks and Financial Institutions:** Regulatory bodies like the Reserve Bank of India (RBI) and Securities and Exchange Board of India (SEBI) have guidelines and teams dedicated to cybersecurity in the financial sector.

8. Private Cybersecurity Firms: Several private cybersecurity companies operate in India, offering services ranging from consulting and risk assessment to incident response and security solutions.

- These organizations collaborate to address cyber threats, enforce cybersecurity laws and regulations, provide guidelines and advisories, conduct awareness programs, and investigate cybercrimes. They play a crucial role in safeguarding digital infrastructure and combating cyber threats in India.
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Case Study

- Checklist for reporting cyber crime at Cyber crime Police Station.
- Checklist for reporting cyber crime online.
- Reporting phishing emails.
- Demonstration of email phishing attack and preventive measures

1. Checklist for Reporting Cybercrime at a Police Station

When reporting a cybercrime in person at a police station, especially in Bengaluru, Karnataka, follow these steps:

1. Document the Incident:

- **Date and Time:** Note when the incident occurred.
- **Nature of the Crime:** Describe the type of cybercrime (e.g., hacking, online fraud, cyberstalking).

2. Gather Evidence:

- **Screenshots:** Capture images of fraudulent messages, emails, or transactions.
Webasha
- **Emails:** Save copies of any suspicious or threatening emails.
Webasha+1Cyber Crime+1
- **Transaction Details:** Record any financial transactions involved in the fraud.
Income Tax India+2Webasha+2Studocu+2
- **Logs:** Keep logs of any unusual online activity or interactions.
Sancharsaathi+2Webasha+2Cyber Crime+2

3. Identify Relevant Jurisdiction:

- Visit the local Cyber Crime Police Station or the nearest police station in your area.
- In Bengaluru, the Cyber Crime Police Station details are:
 - **Address:** [Insert Address Here]
 - **Contact:** [Insert Contact Number Here]
 - **Website:** [Insert Website URL Here]

4. File a First Information Report (FIR):

- Provide a written complaint detailing the incident.SBI General Insurance
- Attach all collected evidence.SBI General Insurance+2Webasha+2Studocu+2
- Obtain a copy of the FIR for your records.

5. Follow Up:

- Request contact information of the investigating officer.
- Stay in touch for updates on the investigation.

2. Checklist for Reporting Cybercrime Online

To report cybercrimes online in India:

1. Gather Evidence:

- Collect all relevant information as mentioned above.Webasha

2. Access the National Cyber Crime Reporting Portal:

- Visit <https://cybercrime.gov.in/>.

3. Choose the Appropriate Option:

- For crimes against women and children, select "Report Women/Child Related Crime."
- For other cybercrimes, select "Report Other Cyber Crime."

4. Register/Login:

- Create an account or log in using your credentials.

5. Fill in the Complaint Details:

- Enter personal information.Indian Cyber Squad+4OCC.gov+4New York Post+4
- Provide detailed information about the incident.
- Upload supporting evidence.NCSC

6. Submit the Complaint:

- After reviewing, submit the complaint.
- Note the reference number for future tracking.SBI General Insurance

7. Helpline Numbers:

- **Cyber Crime Helpline:** 1930The Times of India+3Sancharsaathi+3Cyber Crime+3
- **National Police Helpline:** 112Webasha+5SBI General Insurance+5bcp.karnataka.gov.in+5
- **National Women Helpline:** 181

3. Reporting Phishing Emails

Phishing emails attempt to deceive individuals into revealing sensitive information. To report them:

1. Do Not Engage:

- Avoid responding or clicking on any links.

2. Report to Authorities:

- **Indian Computer Emergency Response Team (CERT-In):**
 - Forward the phishing email to incident@cert-in.org.in.
- **Income Tax Department (for tax-related phishing):**
 - Forward to phishing@incometax.gov.in.

3. Notify Your Organization:

- Inform your company's IT or security department.CrowdStrike

4. Report to Email Service Provider:

- Use the "Report Phishing" feature in your email client.

5. Update Security Measures:

- Change passwords if you suspect any compromise.New York Post
- Enable two-factor authentication where possible.

4. Demonstration of Email Phishing Attack and Preventive Measures

Phishing Attack Demonstration:

1. Crafting the Email:

- An attacker creates an email mimicking a legitimate organization (e.g., a bank).

- The email contains urgent language prompting immediate action. National Government Services Portal+3 New York Post+3 SBI General Insurance+3

2. Malicious Links:

- The email includes a link to a counterfeit website resembling the legitimate one.

3. Data Collection:

- Unsuspecting users enter sensitive information (e.g., login credentials) on the fake site.

4. Exploitation:

- The attacker uses the collected information for fraudulent activities.

Preventive Measures:**1. Verify Sender Information:**

- Check the sender's email address for authenticity.

2. Inspect Links Before Clicking:

- Hover over hyperlinks to view the actual URL.

3. Look for Security Indicators:

- Ensure websites use HTTPS and display a padlock icon.

4. Be Cautious with Attachments:

- Avoid opening unexpected or suspicious attachments.

5. Regularly Update Software:

- Keep your operating system and applications updated to patch vulnerabilities.

6. Educate Yourself and Others:

- Participate in cybersecurity awareness training programs.

7. Use Security Tools:

- Implement email filters and antivirus software to detect malicious

Case Study: A Real-Time Cyber Crime Reporting and Phishing Attack Scenario**Background of the Incident**

Ravi is a small business owner who regularly uses email and online banking for customer payments and supplier transactions. One morning, he received an email that appeared to be from his bank. The email contained the bank's logo, official language and a subject line stating

“Urgent: Account Verification Required to Avoid Suspension.” The message warned that Ravi’s bank account would be blocked within 24 hours unless he verified his details.

Believing the email to be genuine, Ravi clicked the link provided in the email. The link redirected him to a website that looked exactly like his bank’s official login page. He entered his username, password and OTP. Within an hour, Ravi started receiving SMS alerts showing multiple unauthorized fund transfers from his account.

Immediate Actions Taken by the Victim

Realizing that he had been trapped in a phishing attack, Ravi immediately contacted his bank and blocked his debit card and online banking access. He took screenshots of the phishing email, the fake website URL, and the transaction messages. He preserved all digital evidence instead of deleting anything, as he understood the importance of proof for investigation.

Reporting Cyber Crime at the Cyber Crime Police Station

The same day, Ravi visited the nearest Cyber Crime Police Station. He carried his identity proof, bank passbook, transaction statements showing the fraudulent transfers, screenshots of the phishing email, and details of the suspicious link. Ravi submitted a written complaint describing how the phishing email deceived him and resulted in financial loss. The cyber crime officer registered an FIR and assured him that the complaint would be forwarded to the cyber forensics team for further investigation.

Reporting Cyber Crime Online

In addition to visiting the police station, Ravi also reported the incident through the **National Cyber Crime Reporting Portal**. He selected the category **“Online Financial Fraud / Phishing”** and entered his personal details, bank account information, transaction IDs and the amount lost. He uploaded screenshots of the phishing email and transaction receipts. After submitting the complaint, Ravi received an acknowledgment number, which he saved for future follow-up.

Reporting the Phishing Email

Ravi forwarded the phishing email to his bank’s official phishing-report email ID and marked it as “Phishing” in his email account. He also reported the email to the cyber crime reporting portal so that the malicious website could be blocked and other users could be warned. This step helped authorities take preventive action against the attackers.

Demonstration of the Phishing Attack

The investigation revealed how the attack worked. The attackers created a fake email and a fraudulent website that closely resembled the bank's official portal. The email used fear and urgency to force Ravi into acting quickly. Once Ravi entered his credentials, the attackers captured them in real time and transferred money using his account access. The phishing website disappeared soon after the attack to avoid detection.

Preventive Measures Learned from the Case

From this incident, Ravi learned several important lessons. He realized that banks never ask customers to share passwords or OTPs through emails or links. He enabled two-factor authentication for all accounts, regularly checked his bank statements, and educated his employees about identifying phishing emails. He also installed updated security software on his devices and avoided clicking unknown links.

Real-Time Cyber Crime Story: QR Code Phishing and Reporting

In 2024, Meena, a school teacher, wanted to sell her old smartphone through an online marketplace app. Within a few hours of posting the advertisement, she received a call from a person claiming to be a buyer. The caller sounded genuine and spoke politely, saying he wanted to make an advance payment immediately. He sent Meena a **QR code** through WhatsApp and asked her to scan it to "receive" the money.

Unaware that money is never received by scanning a QR code, Meena scanned it and entered her UPI PIN as instructed by the caller. Within seconds, ₹45,000 was debited from her bank account. The caller disconnected the call and blocked her number. Meena immediately realized that she had fallen victim to a **social engineering-based cyber fraud**.

Meena contacted her bank helpline and blocked her UPI account. She took screenshots of the WhatsApp chat, call details, QR code image, bank SMS alerts, and transaction ID. On the same day, she reported the incident on the **National Cyber Crime Reporting Portal** under the category *Online Financial Fraud*. She filled in her personal details, uploaded evidence, and received an acknowledgment number. Later, she visited the nearest Cyber Crime Police Station with her ID proof, bank statement, and written complaint. The cyber crime cell registered the case and coordinated with the bank to trace the fraudulent account.

The investigation revealed that the cyber criminal had used a fake SIM card and mule bank account to withdraw money quickly. This case highlighted how attackers exploit trust, urgency, and lack of awareness rather than technical hacking. Meena later participated in a cyber awareness program and educated others that **QR codes are only for making payments, not receiving money.**

Learning Outcome from the Story

- Cyber criminals now use **simple social engineering techniques**
- Immediate reporting increases chances of fund recovery
- Evidence preservation is critical
- User awareness is the strongest defense

Real-Time Cyber Crime Story: Fake Job Offer and Malware Attack

In early 2024, Arjun, a final-year engineering student, received a message on LinkedIn offering a “Work From Home” job with a reputed multinational company. The message included a professional-looking email ID and company logo, which made the offer appear genuine. The recruiter asked Arjun to complete a simple registration process and download an “offer letter” sent as an email attachment.

Without suspecting anything, Arjun downloaded and opened the attachment. Unknown to him, the file contained **malware** that silently installed itself on his laptop. Over the next few days, Arjun noticed his system becoming unusually slow and observed unauthorized login alerts from his email and bank accounts. Soon, money was transferred from his bank account without his knowledge.

Realizing something was wrong, Arjun immediately disconnected his laptop from the internet and contacted his bank to freeze his account. He took screenshots of the LinkedIn messages, fake recruiter emails, downloaded file details, and bank transaction alerts. Arjun reported the incident on the **National Cyber Crime Reporting Portal** under the category *Online Job Fraud and Malware Attack*. He also visited the Cyber Crime Police Station with his identity proof, system logs, and a written complaint. The cyber cell confirmed that the attachment was a **Trojan malware** used to steal credentials.

The attackers had used fake job postings to target students seeking employment. Through investigation, authorities blocked the fraudulent email domain and warned users through

awareness notices. Arjun later learned the importance of verifying job offers and avoiding unknown attachments.

Learning Outcome from the Story

- Fake job offers are a common **recent cyber fraud trend**
- Malware often spreads through email attachments
- Early detection reduces financial damage
- Reporting helps stop similar attacks on others

Real-Time Cyber Crime Story: WhatsApp Account Takeover Scam

In mid-2024, Suresh, a college lecturer, received a WhatsApp message from a number that appeared to belong to one of his colleagues. The message requested an urgent **6-digit verification code**, claiming it was sent by mistake while setting up a new phone. Trusting the sender, Suresh shared the OTP without verifying.

Within minutes, Suresh was logged out of his WhatsApp account. The cyber criminal had taken complete control of his account and started sending messages to Suresh's contacts asking for money, claiming it was an emergency. Two of his contacts transferred money before realizing the fraud.

When Suresh noticed that he could no longer access WhatsApp, he immediately informed his friends through phone calls and social media. He reported the incident through the **National Cyber Crime Reporting Portal** under the category *Social Media Account Hacking*. He also visited the Cyber Crime Police Station with screenshots of messages, OTP details, and call logs. The cyber crime cell helped him recover the account and trace the fraudulent activity.

The investigation revealed that attackers commonly use **OTP-based social engineering** to hijack messaging accounts. Suresh later enabled two-step verification on WhatsApp and educated his contacts never to share OTPs.

Learning Outcome from the Story

- OTP sharing can lead to full account takeover
- Social engineering targets trust, not technology
- Quick reporting limits damage
- Two-step verification is essential

Real-Time Cyber Crime Story: Fake Courier / Customs Scam

In early 2025, Priya, a postgraduate student, received an automated phone call claiming to be from a courier service. The call stated that a parcel in her name contained illegal items and had been stopped by customs authorities. The message warned that legal action would be taken unless she contacted a given number immediately.

When Priya called the number, a person posing as a customs officer answered and spoke confidently, using official terms and threatening arrest. The caller transferred the call to another person pretending to be a cyber crime officer, who instructed Priya to complete a “verification process” to avoid legal trouble. Under pressure and fear, Priya was asked to transfer money to a “temporary verification account” and share her Aadhaar details.

After transferring ₹62,000, the call disconnected. Priya soon realized it was a scam when the number became unreachable. She immediately informed her bank and blocked further transactions. She collected call recordings, screenshots of payment receipts, and phone numbers used in the scam.

Priya reported the incident on the **National Cyber Crime Reporting Portal** under *Impersonation and Online Financial Fraud*. She also visited the Cyber Crime Police Station with her ID proof and written complaint. The investigation revealed that the attackers used VoIP numbers and fake identities to impersonate officials and extort money.

Learning Outcome from the Story

- Cyber criminals exploit **fear and authority**
- Government agencies never demand money over phone calls
- Immediate reporting increases chances of recovery
- Awareness is the strongest defense

Real-Time Cyber Crime Story: Online Shopping Fraud via WhatsApp and GPay

In late 2024, Lakshmi, a homemaker, saw an advertisement on social media offering branded kitchen appliances at very low prices. The post asked interested buyers to contact a WhatsApp number for orders. When Lakshmi messaged the number, the seller shared attractive product images, fake customer reviews, and promised quick delivery. The seller insisted that payment should be made immediately through **Google Pay (GPay)** to confirm the order.

Trusting the seller, Lakshmi transferred ₹18,500 using GPay and received a confirmation message on WhatsApp. However, after the payment, the seller stopped responding to messages and the WhatsApp number was switched off. The product was never delivered. When Lakshmi checked the social media page, it had already been deleted.

Realizing she was cheated, Lakshmi immediately contacted her bank and raised a transaction dispute. She took screenshots of the WhatsApp chat, GPay transaction details, UPI ID, and the advertisement. She then reported the incident on the **National Cyber Crime Reporting Portal** under the category *Online Shopping Fraud*. Lakshmi also visited the nearest Cyber Crime Police Station with her ID proof and written complaint. The cyber crime cell informed her that such frauds are common, where attackers create fake WhatsApp business accounts and collect money through UPI before disappearing.

Learning Outcome from the Story

- Fraudsters misuse WhatsApp for fake online shopping
- Advance payment through UPI is risky without verification
- Evidence collection is crucial for investigation
- Only trusted and verified shopping platforms should be used

Real-Time Cyber Crime Story: Fake Electricity Bill Payment Link

In 2025, Mohan, a retired government employee, received a WhatsApp message claiming to be from the electricity board. The message stated that his electricity connection would be disconnected within two hours due to non-payment of the bill. It included a payment link and asked him to clear the dues immediately using UPI.

Worried about the disconnection, Mohan clicked the link and was redirected to a fake payment page that closely resembled the official electricity board website. He entered his mobile number and approved a UPI request, believing he was paying the bill. Instead of paying a small bill amount, ₹27,000 was debited from his bank account.

After realizing the fraud, Mohan immediately contacted his bank and blocked further UPI transactions. He saved screenshots of the WhatsApp message, fake payment link, UPI transaction details, and SMS alerts. Mohan reported the incident on the **National Cyber Crime Reporting Portal** under *Online Payment Fraud* and later visited the Cyber Crime Police Station with his ID proof and written complaint.

The cyber crime investigation revealed that attackers commonly send fake utility payment messages to senior citizens, exploiting urgency and fear of service disruption. Mohan later shared his experience in his residential association to spread awareness.

Learning Outcome from the Story

- Fake utility payment messages are a common scam
- Urgency is used to manipulate victims
- Immediate reporting reduces loss
- Official payments should be made only through verified apps

QUESTIONS:

1. Classify cyber crimes with suitable examples.
2. Explain cyber crimes targeting computers and mobile devices.
3. Describe cyber crimes against women and children.
4. Explain common financial frauds in cyberspace.
5. Describe various social engineering attacks with examples.
6. Explain malware and ransomware attacks and their impact.
7. Differentiate between zero-day attacks and zero-click attacks.
8. Explain the modus operandi of cyber criminals.
9. Describe the procedure for reporting cyber crimes in India.
10. Explain remedial and mitigation measures against cyber crimes.
11. Discuss the legal perspective of cyber crime in India.
12. Explain the objectives and key provisions of the IT Act 2000.
13. Analyze how amendments to the IT Act address emerging cyber crimes.
14. A victim loses money due to an online shopping fraud through UPI. Explain the cyber crime involved and reporting process.
15. Evaluate the effectiveness of existing cyber laws in preventing financial frauds.